Attitudes towards artificial intelligence at work: Scale development and validation

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Abstract

Research suggests that understanding workers' attitudes toward artificial intelligence (AI) application is a prerequisite to successfully integrating AI into an organization. However, few studies have clarified the meaning of attitudes towards AI application at work (AAAW) as a multi-faceted construct that can be assessed with psychometric validity. To address this issue, we developed and validated a scale to capture individuals' AAAW using three independent samples (total *N* = 2,841). The resulting 25-item scale covers an overall construct of AAAW as well as six dimensions that are subsumed under the construct (i.e., perceived humanlikeness, perceived adaptability, perceived quality of AI, AI use anxiety, job insecurity, and personal utility). Our findings suggest that the AAAW scale has good psychometric properties and is able to predict important recruiting outcomes. The scale offers opportunities to better understand and measure workers' attitudes towards AI application at work in a comprehensive and integrative manner.

Keywords: artificial intelligence, attitudes towards AI application, worker, measurement development

Attitudes Toward Artificial Intelligence Application at Work:

Scale Development and Validation

Artificial intelligence (AI) is commonly defined as a system's ability to interpret external data, learn from the data, and achieve specific goals through adaptation (Kaplan & Haenlein, 2019). As such, AI is rapidly changing many aspects of work, including job characteristics, interpersonal interactions, workplace decision making, and even organizational structures (e.g., Dietvorst et al., 2019; Kaplan & Haenlein, 2019; Tambe et al., 2019). In adopting AI technology, organizations face varying degrees of reluctance, skepticism, willingness, and/or enthusiasm towards AI technology as experienced and expressed by their employees. Past research has shown that individuals differ in anthropomorphizing (Waytz et al., 2010), emotionally reacting to (Shank et al., 2019), and evaluating the usefulness of AI (Dietvorst et al., 2016), and that such attitudes can hinder or facilitate effective use of AI (Del Giudice et al., 2023; Tambe et al., 2019) and organizational outcomes (Glikson & Woolley, 2020). Therefore, understanding and managing employee attitudes towards AI is a key to successful implementation of the technology.

Despite the importance, however, the literature on this topic remains scarce and fragmented on what specific psychological elements make up such attitudes and how they relate to important work outcomes. The lack of empirical studies may be attributed to the absence of a reliable and valid framework for conceptualizing and measuring the construct. Workers' attitudes towards AI is a multifaceted construct that involve various factors ranging from functionality to socio-emotionality (Park & Woo, 2022), and from appreciative (Logg et al., 2009) to aversive attitudes (Dietvorst et al., 2016). However, existing measures of technology-related attitudes are limited in that they have focused on either functional aspects such as perceived usefulness and ease of use (Davis, 1989), or more socio-emotional issues related to job insecurity (Brougham & Haar, 2018) and trust (Schaefer, 2016), rather than capturing a full range of cognitive and affective elements of attitudes. Moreover, most of the existing scales were developed for a general population and context (i.e., how people think and feel about new technology) rather than focusing on unique characteristics and attitudes associated with AI applications in the workplace setting. To address these concerns, the current article introduces a measure that represents the multidimensional nature of the focal construct of interest, which we refer to as *attitudes towards AI application at work* (AAAW).

The current study makes several theoretical and practical contributions. By integrating multiple aspects of worker attitudes towards AI application, it enables future research to take into account distinct (e.g., social and utility-related) factors that may boost and hinder the adoption of new technology in the workplace (Yuan & Woodman, 2010). Doing so facilitates more nuanced understanding of how workers react to, interact with, and experience different features of the AI technology, which is recognized as a critical area of future research (Del Giudice et al., 2023; Glikson & Woolley, 2020; Tambe et al., 2019).

Also, the multidimensional measurement framework provides opportunities to comparatively evaluate specific aspects of worker's attitudes, which offers more targeted prediction over criteria of interest (e.g., performance, satisfaction, stress/burnout). For example, workers' attitudes related to job insecurity (Brougham & Haar, 2018) may be particularly relevant to stress/burnout and turnover intentions whereas functionality-related attitudes may be more predictive of employee performance.

Worker Attitudes towards AI Application and Existing Measures

In this study, we broadly define AI applications as the utilization of AI technology in the work setting to improve organizational performance. There are various types of AI technologies implemented in today's workplace, such as conversational AI technology that generates interview questions upon interviewers' request during the recruiting and selection process, and a type of cognitive AI that analyzes customers' voice and/or words and generates adaptive responses accordingly. These applications are often implemented at the organizational level and intended to increase productivity, decision-making, and efficiency. AI technologies have unique attributes that differentiate them from other technologies, such as a certain degree of self-learning and autonomy.

We define attitude as “a psychological tendency that is expressed by evaluating a particular entity with some degree of favor or disfavor” (Eagly & Chaiken, 1993). Such evaluation entails cognitive and affective responding including beliefs, thoughts, feelings, and emotions (Eagly & Chaiken., 2007). Although some scholars incorporate behavioral aspects into the definition of attitude, we do not consider behavioral intention as part of the attitude construct. This is in alignment with the perspective that behavioral intention is a consequence of attitudes toward an object, leading to a specific action (Ajzen & Fishbein, 1972).

Understanding individuals’ attitudes towards AI application at work begins with recognizing multiple psychological mechanisms that are interrelated yet distinct, such as affective and cognitive components. Several scholars have developed theories and corresponding measurement tools on attitudes toward new technology and AI. The technology acceptance model (TAM; Davis, 1989) is arguably the most widely used theory for explaining intentions to use technology in various occupational contexts (Venkatesh & Davis, 2000). The TAM posits that one’s intention to use a technology system is determined by two factors: perceived usefulness (i.e., the extent to which an individual believes the technology will improve their performance) and perceived ease of use (i.e., the extent to which an individual believes the technology will be free of effort). Applying the TAM to AI technology within the work context, research has suggested that workers’ perception of AI’s functionality is critical in predicting their willingness to utilize AI in their workplace (e.g., Xu & Wang, 2021).

In addition to these cognitive aspects, research also suggests that people may experience either positive (e.g., happiness, amazement; Shank et al., 2019) or negative (e.g., fear, threat; Liang & Lee, 2017; Shank et al., 2019) emotions associated with new technology. Such emotional responses are distinct from cognitive evaluations of the technology's functionality, but can shape the person's ultimate attitudes toward the technology and whether they intend to use or interact with it in the future (e.g., De Guinea & Markus, 2009; Liang & Lee, 2017). Furthermore, studies have shown that negative emotional reactions to AI at work can have unfavorable impacts on one' career satisfaction and turnover intentions (Brougham & Haar, 2018). Despite the significance of emotionality, however, few studies have examined both cognitive and emotional aspects of attitudes towards AI – possibly due to the lack of an integrative measurement tool that reliably captures these components simultaneously.

Another line of research has highlighted distinctive features of AI that distinguish it from other technology types, such as humanlikeness and autonomous nature. Humanlikeness is perceived in the process of anthropomorphism, in which people attribute to nonhuman objects human characteristics such as motivations, intentions, and emotions (Epley et al., 2007). Anthropomorphism involves attributing human-like mind (e.g., intentions, emotions) and physical features of human to nonhumans (Waytz et al., 2010). The perception of humanlikeness is one of the critical factors in predicting a wide array of attitudes and behaviors towards services and products using AI in business (Glikson & Woolley, 2020; Waytz et al., 2010). These dimensions are unique to the specific domain of AI (vs. general domain of technology), which are not captured in the existing measures of technology-related attitudes.

Although other scales have been developed to measure attitudes towards AI application, they are either focused on a specific aspect of AI (Brougham & Haar, 2018) or broadly concerned with general attitudes towards AI without workplace relevance (e.g., Schepman & Rodway, 2020; Sindermann et al., 2021). For example, Brougham and Haar (2018) developed a new unidimensional measure to assess employee perceptions of whether Smart Technology, Artificial Intelligence, Robotics, and Algorithm (STARA) is likely to impact their future career prospects. The item content of this scale was based on a measure of job insecurity. The other scale by Schepman and Rodway (2020) was developed to assess general attitudes towards AI. This measure is bidimensional and consists of the dimensions of a positive attitude towards AI (e.g., “Artificial Intelligence is exciting”) and a negative attitude towards AI (e.g., “I find Artificial Intelligence sinister”). Even research on the newly developed scales calls for a scale to bring insights into facets of attitudes towards AI (Sindermann et al., 2021).

Our review suggests that there is an increasing body of research examining attitudes towards AI and outputs generated by AI, and new scales are being developed to measure such attitudes. However, findings from the existing studies are difficult to integrate due to their idiosyncratic and fragmented nature, and it remains uncertain whether they could be applied to workplace contexts. The lack of a unified understanding of attitudes towards AI application at work may hinder the progress of this emerging area of research. To fill this void, the present research aims to develop a scale that reflects the multidimensional nature of AI attitudes, with a substantive focus on AI applications in the workplace.

The Current Study

Our scale development and validation effort spans across three studies that involve online survey data from workers from diverse occupational backgrounds in the United States (total N = 2,841). In the first study (Study 1), we generated items and administrated a questionnaire to 830 U.S. workers. Six dimensions emerged from exploratory factor analysis: perceived humanlikeness, perceived adaptability, perceived quality of AI, AI use anxiety, job insecurity, and personal utility. We then reduced the number of items using exploratory and confirmatory factor analysis techniques as recommended by Hinkin (1998). Next, in Study 2, we examined the AAAW scale's validity by evaluating the internal factor structure as well as empirical associations between the (sub)scale scores, psychological needs, and measures of personality such as the Five-Factor Model dimensions of personality, trust propensity, personal innovativeness in IT, and occupational self-efficacy. Lastly, Study 3 examined how the six AAAW subscale scores differentially predicted job applicants' recruiting outcomes, providing empirical evidence for the measures' utility in a context where the impact of AI application is increasingly recognized (Folger et al., 2022; Mirowska & Mesnet, 2022; van Esch et al., 2019). We also investigated convergent and divergent relationships of the AAAW subscales with various attitudinal outcomes (e.g., job satisfaction, organizational commitment).

The current study makes several theoretical and practical contributions. By integrating multiple aspects of worker attitudes towards AI application, it enables future research to take into account distinct (e.g., social and utility-related) factors that may boost and hinder the adoption of new technology in the workplace (Yuan & Woodman, 2010). Doing so facilitates more nuanced understanding of how workers react to, interact with, and experience different features of the AI technology, which is recognized as a critical area of future research (Del Giudice et al., 2023; Glikson & Woolley, 2020; Tambe et al., 2019).

Also, the multidimensional measurement framework provides opportunities to comparatively evaluate specific aspects of worker's attitudes, which offers more targeted prediction over criteria of interest (e.g., performance, satisfaction, stress/burnout). For example, workers' attitudes related to job insecurity (Brougham & Haar, 2018) may be particularly relevant to stress/burnout and turnover intentions whereas functionality-related attitudes may be more predictive of employee performance.

Study 1: Scale Creation and Factor Structure

The goal of Study 1 was to develop an initial set of the AAAW items and identify the underlying factor structure of the AAAW construct. We aimed to develop a scale using an empirically driven approach informed by prior research, rather than a completely deductive approach. We chose to do so given the fact that previous studies in this general domain have generated a number of measurement items that have proven theoretically relevant and empirically useful, yet there was not a single conceptual framework that ties them together in an integrative manner. Our approach to identifying dimensions of the AAAW scale was thus devoid of expectations regarding the specific number of dimensions or hierarchical orders of dimensions. Rather, we started with exploratory factor analysis using existing scales on diverse attitudinal aspects towards AI and AI application, and then explored how the items could be subsumed under a set of dimensions (i.e., factors) of the AAAW construct in light of insights gained from psychological theories of attitudes. The resulting 6-factor solution then led to a greater clarity regarding how the construct of AAAW may be best conceptualized and measured. Our primary aim was to compose a scale that measures distinct and related attitudinal constructs under the broad theme of AAAW. Beyond expecting correlations among the dimensions, we were also open to exploring the scale's hierarchical structure with lower- vs. higher-order factors, which did not turn out to be the case (see the results below for details).

Item Generation

To generate an initial pool of items, we first collected publicly available survey items from the existing literature on technology acceptance, human-computer interaction, and AI characteristics such as anthropomorphism (see Appendix A for a list of references that provided the relevant items and sample items). We restricted our search to peer-reviewed journal articles written in English, but otherwise sought to be as comprehensive as possible in our literature searches. Specifically, we used a computer-based electronic search within PsycInfo for the years 1980 through 2019, using keywords such as artificial intelligence, algorithm, work, and worker. Recognizing the interdisciplinary nature of the subject, we selected nine relevant fields, such as applied psychology, multidisciplinary psychology, management, and ergonomics. We also manually searched articles based on the references.

Items on unique aspects of AI such as humanlikeness and learning ability were adapted from prior research without any modification because these characteristics can be applied across domains. For example, we adapted items such as “AI has ability to experience emotions” (Waytz et al., 2010) and “AI is like a person” (Rijsdijk et al. 2007) from prior research without modification. Besides those cases, most items were modified to make them directly applicable to AI application at work. For example, we modified “I am concerned that robots would be a bad influence on children” (Nomura et al., 2006) into “I am concerned that AI would be a bad influence on workers,” and revised “I am personally worried about my future in my industry due to AI replacing employees” (Brougham & Haar, 2018) into “I am worried about my career due to AI replacing employees.” All the items were carefully reviewed by the first two authors of the current article, serving as subject-matter experts (with extensive knowledge in human-technology interactions and industrial-organizational psychology) for writing clarity, content relevance, and appropriateness of item modification that corresponds to AI and AI usage in the workplace. Through an iterative process of multiple independent reviews and collective discussions, we modified the wording that applies to the workplace and workers and avoided unnecessarily redundant and ambiguous items (DeVellis, 2012; MacKenzie et al., 2011). This process resulted in a pool of 104 items.

Method

Participants and Procedures

We recruited 917 working adults in the United States using Amazon Mechanical

Turk (MTurk). We removed data from 87 individuals who either did not complete the survey or failed three attention check items, leaving a total of 830 participants. The participants were rewarded $1.70 for completing the survey that took approximately 19 minutes on average. The participants were on average 36.15 years old (*SD* = 11.43) ranging from 18 to 78 years, European American (77.23%), and 55.54% identified as female. The participants responded to 104 items (generated from the aforementioned process) on a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*) in addition to questions regarding their demographic characteristics. A 5-point Likert scale was chosen because individuals may not have fully formed their opinions on AI technology, given its relatively novel nature. Including a neutral point rather than forcing to choose "agree" or "disagree" would be meaningful in capturing the nuanced attitudes towards AI.

Analyses

Before conducting exploratory factor analysis (EFA), conformity of the data structure for the factor analysis was examined through Kaiser–Meyer–Olkin (KMO) test and Bartlett's test. The KMO statistic was .96, higher than the recommended .60 threshold, and Bartlett's test of sphericity was significant ($p < .001$), supporting that factor analyses could be applied to this data. Next, we conducted EFA on the 104 items using principal axis extraction with promax rotation because the factors were expected to correlate. Multiple indices were used to decide how many factors to retain including eigenvalues (Kaiser, 1960), scree plots (Cattell, 1966), parallel analysis (Horn, 1965; Humphreys & Montanelli, 1975) and interpretability. We also examined the structure of different rotated solutions (e.g., direct oblimin rotation) to evaluate the conceptual clarity.

Results and Discussion: A Six-Factor Structure

Through exploratory analyses, we found a six-factor solution was interpretable and parsimonious and retained 43 items across the six factors (See Appendix B for details in factor analyses and the results). We labeled the six factors (or dimensions) as *perceived humanlikeness, perceived adaptability, perceived quality, AI use anxiety, job insecurity,* and *personal utility.* The first dimension identified is *perceived humanlikeness* of AI, captured by items such as "AI has desires." This dimension describes the extent to which workers perceive AI technology has humanlike properties such as emotions and free will. The factor reflects properties captured in attributing human uniqueness and higher order cognition to AI, being affected by both physical features and of AI and functions of AI (Waytz et al., 2010).

The second dimension identified is *perceived adaptability* of AI, measured by items such as "AI can learn at work." It concerns the extent to which workers perceive AI technology has capabilities of learning and adaptability. Unlike perceived humanlikeness that links with mental state attribution in general, this factor measures individuals' perception of AI's specific ability to adapt to uncertain situations and improve based on information processing (Kaplan & Haenlein, 2019).

The third dimension is *perceived quality* of AI, measured by items such as "AI produces correct information" and "AI operates reliably." This dimension represents an individual's perceptions about a functional quality of AI and its outcomes regarding reliability, formatting, and correctness of information. The emergence of this factor is in line with prior research suggesting the importance of functional aspects in using new technology (Wixom & Todd, 2005).

The fourth dimension is *AI use anxiety*, marked by items such as "Using AI for work is somewhat intimidating to me." This attitudinal dimension captures a worker's anxious feelings about the practical usage of AI at work. It appears to reflect not only one's general dispositions toward technological innovation (Park & Woo, 2022) but also the environmental characteristics of the workplace (Yuan & Woodman, 2010). In a work context surrounded by others and their appraisals, workers gauge their expected gains and losses before exhibiting innovative behaviors such as using new technology (Yuan & Woodman, 2010). Using and adopting new technology such as AI raises concerns for image gains and losses, and these concerns might help emerge AI use anxiety as an important dimension.

The fifth dimension is *job insecurity* caused by AI, being captured by items such as "I think my job could be replaced by AI." The factor represents an overall concern about the continued existence of a particular job and industry due to possible replacement by AI technology. Regardless of an objective reality, how an individual views AI forms a perception of job insecurity and perceived job insecurity has been commonly observed across jobs and occupations (Brougham & Haar, 2018; Nam, 2019). Job insecurity dimension emerges not merely as one's own job loss in the near future, but as the existence of their job, career, and human's roles in the workplace.

The last dimension is labeled *personal utility*, being captured by items such as "Using AI would allow me to have increased confidence in my skills at work." This factor assesses a perception of usefulness and satisfaction of using AI at work based on perceived value and expectation. The factor reflects how using AI would satisfy workers' confidence, independence, and feelings of enjoyment at work. It is closely related to the construct of perceived usefulness in the TAM theory (Davis, 1989; Venkatesh & Davis, 2000), but the personal utility dimension tends to encompass a broader content by also including positive affective responses towards AI use at work.

These six dimensions tap onto constructs identified by previous research on technology acceptance (De Guinea, & Markus, 2009) and AI characteristics (Rijsdijk et al., 2007; Waytz et al., 2010). At the same time, going beyond what has been identified in past studies, the AAAW measurement framework includes two unique features of AI such as perceived humanlikeness and adaptability. In addition, the AAAW scale captures the functionality aspects of AI more broadly: In contrast to research focusing on specific functionality aspects of technology such as currency, immediacy, format, and accessibility (Rijsdijk et al. 2007; Wixom & Todd, 2005), the AAAW framework groups them into an overarching dimension of perceived quality of AI. Taken as a whole, the AAAW scale covers comprehensive attitudinal aspects towards AI application in the workplace.

According to Lambert and Newman (2022), the definition of a construct needs to specify stability of the construct as well as the low end of the construct, and construct gradation. Our theoretical assumption is that the structure of AAAW is stable across time, culture, and AI types, although the values of each construct may change over time. We also conceptualize that the AAAW dimensions are unipolar: the low end of the dimension would entail an absence of the construct. For example, the low end of perceived humanlikeness presents the AI technology is not perceived as humanlike, and the low end of AI use anxiety indicates that one is not anxious about using AI at work (rather than indicating a sense of joy using AI at work).

The AAAW scale also reflects the characteristics of workers *and* the workplace. One of the dominant reactions to AI is job insecurity caused by a realistic threat such as job loss (Brougham & Haar, 2018; Nam, 2019), which has been absent in the measures toward new technology in general. Interestingly, two affect-laden dimensions emerged from our factor analyses were both negative in nature (i.e., anxiety and insecurity). The finding that workers' negative emotions seem to override positive emotions towards AI application at *work* is at odds with other research showing distinct positive emotional responses towards AI as a *consumer* (Shank et al., 2019). In the workplace, rather than instant positive emotional responses towards AI application, workers' positive attitudes seem to be more closely mixed with cognitive appraisals of AI such as personal utility.

Study 2: Scale Refinement and Validation

We conducted Study 2 with three goals related to improving the scale's overall validity and practical utility. First, we sought to reduce the number of items to minimize response fatigue and to improve psychometric properties, by selecting best-performing items based on EFA (i.e., how well each item captures the underlying theoretical factor structure). Second, we evaluated whether the internal (factor) structure found in Study 1 could be replicated in another sample via confirmatory factor analysis (CFA) as an important piece of validity evidence for the scale (AERA et al., 2014; Hinkin, 1998). If subsets of the items that have specific characteristics in common function differently, this indicates multidimensionality of a scale (AERA et al., 2014). Third, we further evaluated the AAAW scale's validity based on its relations with other (external) variables: personality traits. Specifically, we examined whether the six AAAW subscale scores showed a pattern of convergent and discriminant relationships with scores of Big Five personality traits and psychological needs in a theoretically meaningful way (AREA et al., 2014). We briefly expand on our theoretical expectations for these relationships below.

Drawing from informational-processing perspectives of attitude formation, we use personality traits in assessing the validity of the AAAW scale. The expectancy-value model of attitude posits that attitudes toward an object are formed by accessible beliefs and cognitions that individuals hold about the attitude object (Fishbein & Ajzen, 1975). Those cognitions become available to people through personality traits that offer a systematic lens through which individuals view and make sense of their environment (McAdams & Pals, 2006). Research suggests that broad dimensions of dispositional traits (such as Big Five personality; Costa & McCrae, 1992) and more context-specific personality constructs (sometimes referred to as "motivational and social-cognitive developmental adaptations"; McAdams & Pals, 2006) both contribute to individual differences in attitudes towards AI technology (Park & Woo, 2022).

Specifically, we expect that *AI use anxiety* would be positively related to neuroticism.

Neuroticism represents a tendency to express psychological distress (Costa & McCrae, 1992), and those high in neuroticism tend to resist to new changes and adoption of new technology (Barnett et al., 2015). *Job insecurity* would be also positively related to neuroticism because those with high neuroticism are predisposed to negative affective states and social presence at workplaces might exacerbate the negative affective states (Uziel, 2007) in using AI technology at work. *Personal utility* would have positive associations with extraversion, conscientiousness, and openness. People with high extraversion are predisposed to positive orientation and tend to perform better under social presence (Uziel, 2007), suggesting that they would be more likely to feel satisfied and competent in using AI in the workplace. Conscientious people are goal-orientated and they may view functional aspects of AI as an opportunity to improve their performance (Park & Woo, 2022) and facilitate their goal progress, thus they likely exhibit high levels of personal utility. Individuals high on openness are intellectually curious and nondogmatic in values and attitudes (Costa & McCrae, 1992). Their tendency to explore novel activities contribute to shaping favorable attitudes toward an unfamiliar agent (Saef et al., 2019) and new technology such as AI (Park & Woo, 2022). Regarding perceived humanlikeness, adaptability and quality dimensions, we expect they would show negligible relations with the Big Five traits because these attitudes would be more influenced by values, prior knowledge and experiences on AI rather than the traits.

In addition to the Big Five personality traits, we examined another trait-like personality variable: trust propensity. Trust propensity refers to a dispositional willingness to rely on others. Building trust is one of critical elements in predicting AI technology usage (Glikson & Woolley, 2020); those who tend to trust others in general are likely to view AI as trustworthy and favorable. Trust allows individuals to rely on technology partly based on unrealistically optimistic beliefs regarding ability and functionality (Dzindolet et al., 2003). Despite the vulnerability of using AI, those with high trust propensity would evaluate AI positively and believe a high chance of positive outcomes of AI. Thus, we predict that trust propensity would be positively associated with *perceived humanlikeness, adaptability, quality* and *personal utility*.

Next, we examined two additional personality variables that are more domain-specific (i.e., personal innovativeness in information technology [PIIT] and occupational self-efficacy). PIIT refers to the extent to which individuals are willing to try out new information technology (Agarwal & Prasad, 1998). Those with high PIIT tend to actively adopt new technology such as AI and show more confidence in using new technology; they exhibit positive attitudes toward emotional, functional, and social aspects of AI based on their knowledge and experiences (Park & Woo, 2022). We predicted that PIIT would be related to all the dimensions of the AAAW in a desirable way; it would be positively associated with perceived humanlikeness, perceived adaptability, perceived quality, personal utility of AI and it would be negatively associated with AI use anxiety and job insecurity.

Occupational self-efficacy refers to the confidence a person has concerning the ability to fulfill the task at work (Rigotti et al., 2008), which serves as a personal resource that enables one to view new challenges at workplace (e.g., having to learn and use a new technology) in a generally positive light. Those with high self-efficacy tend to undertake new challenges based on positive outcome expectations. In adopting technology, those with high self-efficacy are more likely to form positive perceptions about the use of technology and expect pleasure from using the technology (Kulviwat et al., 2014). Occupational self-efficacy would be related to low levels of AI use anxiety, job insecurity but a high level of personal utility.

Lastly, we explored the associations between ten distinct psychological needs and the six dimensions of the AAAW. Individuals' unmet needs elicit goal-directed perceptions and individuals differ in preferences of psychological needs (Sheldon et al., 2001). Although certain needs may influence attitudes toward a specific facet of attitude towards AI, attitudes towards AI at work are discriminable from psychological needs toward the aspect. For example, regardless of an individual's own competence need satisfaction levels, one might perceive functionality of AI high or low based on their prior knowledge of and familiarity with AI (Gonzalez et al., 2022; Zhang & Dafoe, 2019). Thus, we expect that psychological needs are distinct from the six factors of the AAAW, and they would have weak or null associations.

Method

Participants

We recruited 1,100 adults working and residing in the United States from MTurk. We removed data from 31 individuals who neither completed the survey nor failed three attention check items (e.g., Answer Strongly Disagree for this question), resulting in a total sample size of 1,069 participants. The participants were rewarded $2.00 for completing the survey. Participants were, on average, 39.02 years of age (SD = 12.01) with a range from 18 to 75, female (52.66%), and European Americans (68.21%).

Measures

AAAW. The multidimensional construct of AAAW was measured using 43 items developed and retained from Study 1. The items were rated on a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*) and the scale showed good internal consistency (Cronbach's alphas = .88–.95).

Personality. The Big Five personality traits were measured with 30 items (Soto & John, 2017). The five traits (i.e., extraversion, agreeableness, conscientiousness, neuroticism, and openness) were measured with six items each, and a sample item is "I am someone who is dominant, acts as a leader (extraversion)." *Trust propensity* was measured with eight items (Mayer & Davis, 1999). A sample item includes "Most salespeople are honest in describing their products.” *PIIT* was measured with four items (Agarwal & Prasad, 1998). A sample item includes “I like to experiment with new information technologies.” *Occupational self-efficacy* was measured with six items (Rigotti et al., 2008). A sample item includes “I meet the goals that I set for myself in my job.” These items were rated on a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*) and all of these scales showed acceptable internal consistency (Cronbach’s alphas = .72–.84).

Psychological Needs. Ten distinct psychological needs identified by prior research were measured with 30 items (Sheldon et al., 2001), which consists of three items for each of the ten needs: autonomy, competence, relatedness, self-actualization, physical thriving, pleasure-stimulation, money-luxury, security, self-esteem, and popularity-influence needs. Participants were asked to consider past year of their life and think of the important occurrences during the period. Then they were asked to rate agreement with each item starting with “During this event, I felt...” A sample item is “Very capable in what I did (competence)”. The 30 items were measured on a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*), and all the scales showed acceptable internal consistency (αs = .72–.84) except security need (α = .62) and pleasure need (α = .69).

Scale Refinement

We conducted EFA using principal axis factoring with promax rotation, allowing for correlations among factors and identifying factors based on eigenvalues, scree plot, and parallel analysis. To reduce items, we invited two industrial-organizational psychology researchers who were not involved in item development to review the AAAW items. They were asked to review whether (a) the items were mapped onto the six dimensions; (b) each item was expressed clearly; and (c) each item was nonredundant with other items. Based on their comments and factor analyses, the authorship engaged in multiple iterative discussions to determine which items should be retained. Specifically, we removed items that were deemed conceptually redundant and/or less relevant to the work context. For example, "The information provided by AI is well laid out" was dropped due to conceptual redundancy with "The information provided by AI is well formatted." Also, we agreed that standing in front of AI might not be common at workplace, thus we excluded the item, "I would feel very nervous just standing in front of AI at work."

Ultimately, the process resulted in 25 items that correspond to the six dimensions (see Table 1). The factor analysis of the 25 items also supported the six-factor solution, which accounted for 74.89% of the total variance. The 25-item scale of the AAAW showed good reliability estimates ranging from .84 to .90. The corrected item-total correlations ranged from .62 to .86, and inter-item correlation matrix revealed that all of the items under a factor were positively correlated with means ranging from .52 to .77[1]. The results indicated that high levels of consistencies within each factor, and items under a factor were positively associated.

Results and Discussion

Internal Structure

Using the shortened, 25-item version of the scale, we further evaluated the validity of the 6-factor structure via CFA. In addition to the hypothesized 6-factor solution, we evaluated seven alternative models. The following fit indices were used for model evaluations: chi-square/*df* ratio (χ^2/df), relative fit indices including the comparative fit index (CFI), the Tucker–Lewis index (TLI), the normed fit index (NFI), parsimony-adjusted measures such as root mean square error of approximation (RMSEA), standardized root mean square residual (SRMR), Akaike information criterion (AIC), and Bayesian information criterion (BIC). For

[1]Although all the six factors showed good internal consistency, the items for *perceived quality* showed weaker intercorrelations (.52) than others, which appears to reflect its broad content suggested and observed in prior research (Wixom & Todd, 2005). In developing the AAAW, we aimed to capture a worker's multidimensional attitude towards AI application in a comprehensive manner and develop a scale balancing the depth and the breadth of the content coverage.

χ^2/df, a score of 3 or less indicates a good fit (Kline, 2011). CFI, TLI, and NFI values greater than .95, RMSEA values less than .06, SRMR values less than .05, and lower values AIC and BIC indicate a good fit (Hu & Bentler, 1999; Kline, 2010).

Table 3 presents the results of the CFAs on the seven tested models. The correlated six-factor model showed best fit to the data (Model 8) and the model with two second-order factors (Model 6) also showed relatively good fit indices. The difference between the models was significant ($\chi^2(8) = 602.43$, $p < .001$), showing Model 8 fits better to the data. Examination of AIC values and PNFI values also revealed that the AIC value (i.e., 803.60) for Model 8 was better than that of Model 6 (i.e., 1,390.03), and PNFI values (.83) were identical. Overall results suggest that Model 8 fits best for the data and support the six-factor internal structure of the AAAW.

Parameter estimates of the correlated six-factor model further provided the internal convergent validity evidence. Average variance extracts (AVE) values of the six factors ranged from .52 to .77, which exceeded the required value of .50 (MacKenzie et al., 2011). The composite reliability (CR) values of the six factors ranged from .70 to .94, reaching and exceeding the cutoff value of .70 (Malhotra & Dash, 2011). Also, the factor loadings of the latent factors ranged from .72 to .91. These results indicate that items within a dimension are correlated and coherent.

We further examined discriminant validity evidence among the six factors. Correlations among the six factors showed null to moderate relations ranging from .06 to .62. All the associations among six factors were significant except the two relations: job security's relations with perceived adaptability ($r = .06$) and with personal utility ($r = .06$) were not significant. The strongest correlation was found on the link between perceived quality and personal utility ($r = .62$), suggesting the two functionality-related attitudes are closely related. The findings that the correlations between the six factors were smaller the square root of the

AVE value of each factor (Fornell & Larcker, 1981) provide support for discriminant evidence among the six factors.

Relations with Other Variables

Table 4 presents descriptive statistics and bivariate correlations between the AAAW and study variables. Overall, the patterns of bivariate correlations between the AAAW and the variables were consistent with our theoretical explanations. As expected, neuroticism showed positive associations with *AI use anxiety* and *job insecurity*. *Personal utility* dimension had expected positive associations with extraversion and openness, but the hypothesized relation with conscientiousness was not found. Also, *perceived adaptability* and *perceived quality* had weak to null associations with the five traits as expected.

Trust propensity had positive associations with *perceived humanlikeness, perceived adaptability, perceived quality,* and *personal utility*, but its associations with two negative emotional attitudes such as *AI use anxiety* and *job insecurity* were not significant. Trust propensity seems to play positive roles in workers' attitudes towards AI application by increasing positive attitudes rather than inhibiting negative emotional response towards AI application. PIIT exhibited expected positive roles with all the six dimensions except *perceived humanlikeness*. In line with prior research that supports desirable roles of PIIT in accepting new technology including AI (Oostrom et al., 2013; Park & Woo, 2022), those with high PIIT consistently exhibited positive attitude across attitudinal aspects. Again, expected associations between the AAAW and occupational self-efficacy were observed. Occupational self-efficacy was negatively correlated with *AI use anxiety*, *job insecurity*, and positively correlated with *personal utility*.

Lastly, the correlations between AAAW factors and psychological needs were either small or null. Certain types of needs did not necessarily exhibit strong associations with attitudes relevant to the needs. For example, *personal utility* was weakly correlated with need for competence. Interestingly, *perceived humanlikeness* had relatively strong associations with extrinsic needs such as needs for money and physical striving compared to the other dimensions of the AAAW. The findings echoed by past research on a positive association between materialism and anthropomorphism (Kim & Kramer, 2015). Overall findings were consistent with our expectations.

However, we also observed some unexpected correlations. For example, we found that *perceived quality* and *personal utility* had positive correlations with some psychological needs that are extrinsic in nature, such as need for money, popularity, and physical striving, whereas more intrinsic needs (e.g., autonomy, competence, and relatedness) showed weak or null correlations with the AAAW dimensions as we expected. Individuals driven by extrinsic needs tend to engage in interpersonal comparison and strive for success to establish their self-worth, whereas those with intrinsic needs genuinely seek to develop their competence and foster affiliation with others (Deci & Ryan, 2000). Individuals with extrinsic needs may externalize AI technology to fulfill their unmet needs. By utilizing AI, they expect an easier fulfillment of their extrinsic needs, which shape positive perceptions of perceived humanlikeness and functionality aspects of AAAW.

We also found that *AI use anxiety* and *job insecurity* were negatively associated with agreeableness and conscientiousness. We suspect that the tendencies to maintain harmony with humans and prioritize human ability might buffer negative reactions towards AI. Interesting and unexpected correlations were observed regarding *perceived humanlikeness*; the dimension was negatively related to agreeableness, conscientiousness, and occupational self-efficacy. Confidence in one's competence or tendency to prioritize ability might negatively affect perceived humanlikeness. Those with high self-efficacy or conscientiousness have a desire to gain mastery over the work environment (Boyce et al., 2010; Rigotti et al., 2008), and they might be less willing to attribute human characteristics to technology that could be a threat to human identity. Taken together, the findings showed that the six AAAW subscale scores had a pattern of convergent and discriminant relationships with personality traits.

Study 3: Criteria-Related Validity Evidence Predicting Recruiting Outcomes

In Study 3, we examined additional strands of validity evidence to provide a more coherent and refined interpretation of the AAAW scale. First, we evaluated the criterion-related validity evidence regarding recruiting outcomes – i.e., whether and how the AAAW dimensions predict recruiting outcomes using vignettes that describe recruiting processes that incorporate AI applications. In addition, we further investigated empirical relations of the AAAW scale and its subscales with (a) two existing measures of attitudinal constructs related to AI and interactive robots (i.e., General Attitude toward Artificial Intelligence Scale [Schepman & Rodway, 2020] and the Negative Attitude toward Robots Scale [Nomura et al., 2009]) and (b) three attitude constructs that are not theoretically related to the AAAW constructs (i.e., job satisfaction, organizational commitment, and life satisfaction).

Job Applicants' AAAW and Recruiting Outcomes

Job applicants collect and evaluate information that they use to form organizational perceptions and attitudes during the recruitment and selection processes (Lievens, & Slaughter, 2016). According to a signaling theory perspective (Connelly et al., 2011), applicants use the information they collect and receive about an organization as indicators of organizational characteristics (Bangerter et al., 2012; Folgers et al., 2021). Technologies used in selection processes send signals about a hiring organization during the pre-entry phase. The uptake of AI technology conveys diverse information such as the organization's working conditions (i.e., technology-advanced environment), the capability of an organization (van Esch et al., 2019), and fair perception toward an organization (Folger et al., 2022). Such information shapes an organizational image and affects job applicants' attitudes such as organizational attractiveness (Folger et al., 2022).

Importantly, the effects of AI application during the recruiting process differ depending on job applicants' existing attitudes towards AI application at work. AI technology can be interpreted as positive or negative depending on individuals' attitudes towards AI, which reflect their needs, values, and preferences toward the technology (van Esch et al., 2019; Waytz et al., 2010). For example, the likelihood to accept a job increases when the recruiting system and process convey values that align with job applicants' characteristics (Bretz Jr. & Judge, 1994). Individuals who value innovation and trendiness may favor workplaces with heavy reliance on AI application (van Esch & Black, 2019), whereas individuals who value human involvement and mistrust algorithmic decisions are less likely to find AI-enabled recruiting process to be desirable (Folger et al., 2022; Mirowska & Mesnet, 2022). Applying these ideas to the AAAW framework, we hypothesize that the following four AAAW dimensions would affect the effects of AI application in the recruiting process on recruiting outcomes: perceived quality, personal utility, AI use anxiety, and job insecurity.

Perceived quality of AI would facilitate positive effects of AI application on recruiting outcomes. Organizations that adopt AI technology demonstrate their capability to reinvent products and services, which are critical cues to signal organizational investments (Lievens & Slaughter, 2016). Substantive organizational investments such as adoption of new technology are one of critical cues that affect job applicants' attitudes toward organization (Lievens & Slaughter, 2016). In particular, job applicants' inferences of innovativeness and power based on symbolic and instrumental attributes associated with positive organizational image and recruiting outcomes (Folger et al., 2022; van Esch et al., 2019). Job applicants' positive impressions of a hiring organization partly stem from the effective functions of the recruiting system (Folger et al., 2022) and job applications who perceive good quality of AI technology would evaluate the use of AI during recruiting process as efficient and dependable. This likely contributes to positive recruiting outcomes for those with high perceived quality of AI.

Hypothesis 1: The effects of AI application on recruiting outcomes (i.e., organizational attractiveness, intention to accept job offer, and intention to recommend) are more positive for those who have a high perception of the quality of AI than those with a low perception.

In contrast, *AI use anxiety* might blight the effects of AI application on recruiting outcomes. Anxiety directs one's attention away from the current task and is associated with avoidance-oriented behavior as a way to escape negative stimuli (Eysenck et al., 2007). Specifically, anxiety interferes with self-regulation and interview ability during selection processes, as anxiety exhausts job applicants' resources and thus leads to poor recruiting outcomes (McCarthy et al., 2017). Although direct evidence on AI use anxiety during recruiting process is scarce, job applicants' anxiety related to technology (e.g., technology use anxiety) has shown negative influences on the effects of the AI-enabled recruiting process on job applicants' application likelihood (van Esch et al., 2019).

Hypothesis 2: The effects of AI application on recruiting outcomes (i.e., organizational attractiveness, intention to accept job offer, and intention to recommend) are less positive when the level of AI use anxiety is high than low.

Job insecurity caused by AI may also damage the effects of AI application on recruiting outcomes. The inherent uncertainty in job insecurity prevents individuals from effectively using coping strategies (Lazarus & Folkman, 1984). Job insecurity due to AI replacement doubts the existence of industry and organizations as well as jobs (Nam, 2019). Furthermore, job insecurity caused by AI adversely relates to an array of employee's attitudes such as organizational commitment and intention to stay (Brougham & Haar, 2018; Nam, 2019). Recruiting processes incorporating AI technology would raise concerns regarding job security. Particularly, job applicants with high job insecurity are more likely to experience stress and show reduced engagement during the process. Thus, job insecurity caused by AI would negatively moderate the effects of applying AI on recruiting outcomes.

Hypothesis 3: The effects of AI application on recruiting outcomes (i.e., organizational attractiveness, intention to accept job offer, and intention to recommend) are less positive when the level of job insecurity is high than low.

Personal utility is likely to yield positive moderating effects on the relationship between AI application and recruiting outcomes. Personal utility entails positive emotions and cognitive evaluations associated with AI usage. Job applicants who perceive the system as personally useful will likely develop positive attitudes toward the organization. Moreover, recruiting system features such as website designs often invoke a positive emotional response, and the initial affective response toward the system influences subsequent beliefs and attitudes about the system and organization (Cober et al., 2003). For example, the initial positive emotional experience of job applicants who feel satisfied with an AI-enabled recruiting system may lead to an overall positive impression toward the recruiting process and the hiring company by extension. Thus, the effects of applying AI on recruiting outcomes would be more positive for job applicants with high AI use than those with low personal utility.

Hypothesis 4: The effects of AI application on recruiting outcomes (i.e., organizational attractiveness, intention to accept job offer, and intention to recommend) are more positive for those with a higher perception of personal utility than low.

Method

Procedures

To examine the moderating effects of AAAW on the relationship between AI application and recruiting outcomes, we used vignettes that capture two levels of AI application during the recruiting process. We adopted an example of a “company introduction” text from previous research (Bauer et al., 2001) and modified it to reflect the current job market (e.g., *US News & World Report* was changed to *Glassdoor*). Objective factors such as pay, promotion, prestige, and location, which can influence job applicants’ attitudes, were kept constant across conditions.

Before reading the vignettes, participants were asked to rate their attitudes towards AI application using the AAAW scale. Then, they were randomly assigned to one of the two conditions (i.e., low vs. high AI application conditions). Specifically, we presented descriptions of low versus high levels of AI application during the recruiting process after providing an explicit definition of AI application within the context (Kaplan & Haenlein, 2019). For example, in the high level of AI application condition, we described, “AI conducts interviews and hiring decisions are made by AI and human recruiting teams together.” In the low level of AI application condition, we described “AI transfers applicants’ data to human recruiters and notifies applicants’ results when human recruiting teams make decisions” (See Appendix C for the details of vignettes).

After reading these vignettes, participants were asked to answer two manipulation check items such as “I think that AI technology is widely applied to the recruiting procedures of X Corporation.” Two items (α = .94) were rated on a 7-point Likert scale (1 = *strongly disagree* to 7 = *strongly agree*). The last manipulation item was to ask choose who makes a hiring decision among three categories: AI, human, and AI and human. After that, recruiting outcomes (see *Measures*) were measured. Finally, to examine convergent and divergent validity of the AAAW, they responded to questions related to attitudes towards AI (Schepman & Rodway, 2020) and robots (Nomura et al., 2009), and three attitudinal outcomes (i.e., job satisfaction, organizational commitment, and life satisfaction).

Participants

We recruited 1,002 working adults working in the United States from MTurk. We removed data from 60 individuals who either did not complete the survey or failed three attention check items (e.g., “I was born on February 30”), resulting in a total of 942 participants.

They were rewarded $2.00 for completing the survey. Participants were, on average, 38.65 years of age (*SD* = 11.39) with a range from 18 to 80, female (54.75%), and European Americans (68.21%). Out of 942 participants, 834 participants were currently employed.

Measures

AAAW. The multidimensional construct of AAAW was measured using 25 items retained from Study 2. The items were rated on a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*) and it showed good internal consistency (αs =.86–.94).

Recruiting Outcomes. *Organizational attractiveness* was measured with three items (Crant & Bateman, 1993). A sample item includes “I have a favorable impression of X Corporation.” *Intention to apply* was measured with two items (Robertson et al., 2005). A sample item includes “If I were searching for a job, I would apply to X Corporation.” *Recommendation intention* was assessed with two items (Crant & Bateman, 1993) such as “I would recommend X Corporation to my friends.” These items were assessed on a 5-point Likert scale (1= *not at all*, 5 = *very much*) and they showed good internal consistency (αs = .86–.94).

Other Attitudinal Measures Related to AI and Robots. Two existing measures include 20 items of General Attitudes toward Artificial Intelligence Scale (GAAIS, Schepman & Rodway, 2020) and 14 items of Negative Attitudes toward Robot (NARS, Nomura et al., 2006). A sample item of GAAIS includes “Artificial Intelligence is exciting” and a sample item of NARS includes “I would feel paranoid talking with a robot.” These items were rated on a 5-point Likert scale (1= *not at all*, 5 = *very much*) and they showed good internal consistency (αs = .87–.91).

Job Attitudes. *Job satisfaction* was measured with five items (Judge et al., 2000) and *organizational affective commitment* was assessed with six items (Meyer et al., 1993). A sample item for job satisfaction includes "I feel fairly satisfied with my present job" and a sample item of organizational commitment includes "I enjoy discussing my organization with people outside it." *Life satisfaction* was assessed with five items (Diener et al., 1985). A sample item includes, "The conditions of my life are excellent." These items were rated on a 5-point Likert scale (1= *not at all*, 5 = *very much*), and they all showed good internal consistency (Cronbach's alphas = .90–.91).

Analysis

We first checked whether the vignettes worked well and compared recruiting outcomes between the two conditions using t-test. Then, we examined hypothesized moderating effects of the AAAW attitudes using hierarchical regression analyses. We coded the low level of AI application condition as 0 and the high-level condition as 1, and made interaction terms of the AAAW factors with the experimental condition after centering a moderator (i.e., AAAW dimensions). In the first step, we entered an experiment condition and a moderator and in the second step, we entered the interaction term. Using simple slope analysis at one standard deviation above and below the mean (Aiken & West, 1991), we then further investigated patterns of the interactions as described below.

Results and Discussion

AAAW and Recruiting Outcomes

Before testing hypotheses, we examined gender and age differences between the two conditions. Age did not differ between the high (M = 38.41, SD = 11.28) and low AI (M = 38.86, SD = 11.49) application condition ($t(940) = –0.61$, p = .54). Gender ratio also did not differ between the high AI condition (50.67% female) and the low AI condition (57.89% female; $\chi^2(2) = 5.08$).Then, we examined the effectiveness of the scenario. As expected, participants in the high level of AI application condition ($M = 6.16$, $SD = .81$) perceived the X Corporation adopted AI technology more widely than those in the low AI application condition ($M = 5.01$, $SD = 1.40$; $t(940) = 15.23$, $p < .001$). Also, 89.47% of participants ($n = 442$) in the low AI application condition chose a human as a decision maker during the recruiting process while 1% of participants ($n = 4$) in the high AI application chose a human ($\chi^2(2) = 766.07$, $p < .01$).

We then examined whether the recruiting outcomes differed between the low and high conditions. Job applicants' overall ratings of organizational attractiveness ($M_{\text{low}} = 3.95$, $SD_{\text{low}} = .77$, vs. $M_{\text{high}} = 3.88$, $SD_{\text{high}} = .87$; $t(940) = 1.34$, $p = .181$), intention to apply ($M_{\text{low}} = 4.05$, $SD_{\text{low}} = .81$, vs. $M_{\text{high}} = 4.03$, $SD_{\text{high}} = .88$; $t(940) = .28$, $p = .781$), and intention to recommend the organization ($M_{\text{low}} = 3.69$, $SD_{\text{low}} = .85$, vs. $M_{\text{high}} = 3.62$, $SD_{\text{high}} = .91$; $t(940) = 1.21$, $p = .225$), did not differ as a function of AI application level.

To examine our hypotheses, we proceeded with a series of hierarchical regression analyses. Detailed results are presented in Table 5. Hypothesis 1 predicted *perceived quality* would moderate the effects of AI application on the recruiting outcomes. The interaction between perceived quality and AI application was significant, and added 1% of variances in explaining organizational attractiveness, $F(1, 938) = 7.14$, $p = .007$, and intention to recommend, $F(1, 938) = 7.30$, $p = .007$. It added a smaller (0.4%) variance in explaining intention to apply, $F(1, 938) = 4.31$, $p = .038$. Based on simple slope analysis, we found that the effect of AI application was negative among those with low perceived quality ($B = –.20$, $p < .01$) in predicting organizational attractiveness, while it was not statistically significant for those with high perceived quality. The moderating effect of perceived quality on intention to recommend also showed a similar pattern: the effect of AI application was negative for those with low perceived quality ($B = –.20$, $p < .01$) while its effect was not significant among those with high perceived quality.

Regarding Hypothesis 2, the interaction between AI application and *AI use anxiety* predicted organizational attractiveness with 1% of variance accounted for. As expected, the effect of AI application on organizational attractiveness was negative when the level of AI use anxiety was high ($B = -.24$, $p < .01$), but the effect of AI application on organizational attractiveness was not significant when AI use anxiety was low. The moderating effects on intention to apply and intention to recommend the organization were not statistically significant. Hypothesis 3 that predicted the moderating roles of *job insecurity* was not supported: The interaction effect between AI application levels and job insecurity did not predict any of the recruiting outcomes.

Consistent with Hypothesis 4, the interaction between AI application level and *personal utility* predicted all three recruiting outcomes: organizational attractiveness, $F(1, 938) = 8.36$, $p < .01$, intention to apply, $F(1, 938) = 6.45$, $p < .01$, and intention to recommend, $F(1, 938) = 12.55$, $p < .01$. Similar to results related to Hypothesis 1, however, the pattern of the interaction effects was slightly different from our prediction (although not contrary to the theoretical ideas behind it). The effects of AI applications on organizational attractiveness ($B = –.22$, $p < .01$), intention to apply ($B = –.15$, $p < .01$) and intention to recommend ($B = –.26$, $p < .01$) were all negative among those with low personal utility, whereas these effects were not significant when personal utility was high.

Although we did not hypothesize moderating effects of perceived humanlikeness and perceived adaptability, we explored their effects to extend our understanding of the AAAW scale. *Perceived humanlikeness* had a weak moderating effect on intention to recommend, $F(1, 938) = 4.07$, $p = .044$, $\Delta R^2 = .004$, but its moderating effects on organizational attractiveness and intention to apply were not significant. The patterns of interaction were similar with Hypotheses 1 and 4. The effect of AI application on organizational attractiveness was negative for those with low perceived humanlikeness ($B = -.19$, $p < .05$) while its effect was not significant for those with high perceived humanlikeness ($B = .04$, $p = .60$). *Perceived adaptability* also showed similar results. When perceived adaptability was low, the AI application was negatively associated with organizational attractiveness ($B = -.23$, $p < .01$). However, when perceived adaptability was high, the interaction term did not influence any of the recruiting outcomes.

Overall, the results showed that workers' attitudes towards AI application do add unique variances in explaining the recruiting outcomes. Five AAAW dimensions (except job insecurity) were found to have moderating effects in a theoretically meaningful manner. The moderating effects of personal utility appeared to be the strongest and most robust across all recruiting outcomes. At the same time, we did not found support for moderating effects of job insecurity. Unlike prior research suggesting adverse effects of job insecurity on workers (Brougham & Haar, 2018; Nam, 2019), the negative effects of job insecurity might not be so concerning for (potential) workers at least at the recruiting stage. Future research should clarify why and when that is the case by exploring other potential contingencies.

Despite the significant moderating effects we found, it is worth noting that the observed patterns of the interactions were different from our specific expectations. We had predicted high levels (vs. low levels) of AI application would yield positive recruiting outcomes when perceived quality and personal utility are high. However, consistent patterns were that AI application impacted recruiting outcomes only when workers exhibit low levels of the attitudes. Regarding negative emotional attitudes such as job insecurity and AI use anxiety, the expected patterns were found: the level of AI application decrease recruiting outcomes when participants exhibit high levels of job insecurity and AI use anxiety. The findings indicate that adverse effects of holding negative (vs. positive) attitudes towards AI appeared to be particularly impactful in a recruitment setting. Therefore, rather than focusing on promoting positive attitudes towards AI, cautious efforts to decrease negative attitudes may be more beneficial in improving recruiting outcomes.

AAAW and Workers' Life and Work Attitudes

Discriminant validity evidence in relation to job satisfaction, organizational commitment, and life satisfaction was examined. As expected, the results (Table 6) showed that the AAAW dimensions had null to weak correlations ($|r|$ = .02~ .21) with the three variables. Among the six dimensions, job insecurity had significant and relatively stronger negative associations with job satisfaction (r = –.17), organizational commitment (r = –.17), and life satisfaction (r = –.21) compared to the other dimensions. Although the associations were weak, these results showed that job insecurity caused by AI seem to have wide-ranging adverse effects on life and work attitudes, in line with prior findings (Brougham & Haar, 2018; Nam, 2019). Overall, the weak correlations among the AAAW dimensions and the three attitudes suggest that the six dimensions are discriminant from general attitudes toward job, organization, and life.

The relations between the AAAW and the two existing scales such as GAAIS and NARS provided further convergent and divergent validity evidence. Positive GAAIS (indicating general positive attitudes towards AI) was positively and strongly correlated with personal utility (r = .72) and negatively with AI use anxiety (r = –.59). The other dimensions showed weak to moderate associations with GAAIS (r = .08–.50). Negative GAAIS showed a strong and positive correlation with AI use anxiety (r = .67) and weak to moderate correlations with the other dimensions of the AAAW ($|r|$ = .08 ~ .45). Also as expected, NARS (indicating negative emotional attitudes toward robots; Nomura et al., 2009) had a strong positive correlation with AI use anxiety (r = .74) while only moderately correlated with the job insecurity (r = .27). The other AAAW dimensions also showed weak to moderate correlations with NARS (*r* = -.57 – -.09). The AI use anxiety dimension converged with NARS, while the other dimensions of the AAAW diverged from NARS. The results on the two existing measures suggest that the AAAW captures relatively comprehensive aspects of worker's attitudes towards AI compared with NARS and GAAAIS.

General Discussion

Theoretical Implications

The present study presents a newly developed multidimensional measure of the AAAW to advance the conceptual and empirical understanding of workers' attitudes towards AI application in the workplace. The 25-item scale of the six dimensions (i.e., perceived humanlikeness, perceived adaptability, perceived quality of AI, AI use anxiety, job insecurity, and personal utility) captures a comprehensive range of attitudinal dimensions regarding how workers think and feel about AI application in the workplace. The six-dimensional structure was replicated across two samples, with each dimension showing theoretically expected associations with external variables such as personality characteristics and other measures of attitudes toward new technology. We also found (from Study 3) that some of the AAAW dimensions meaningfully moderate the effects of the AI application on recruiting outcomes.

Among the six AAAW dimensions, three dimensions (i.e., perceived adaptability, perceived quality, personal utility) include functional evaluation of AI technology, which all capture distinct aspects from one another. While perceived adaptability reflects a worker's general perception of AI's specific adaptive function, perceived quality concerns with perceptions of outputs generated by AI at work, and personal utility entails affective (as well as cognitive) experiences a person has regarding AI (e.g., feelings of enjoyment). In predicting recruiting outcomes, the personal utility dimension seemingly has the highest influence on recruiting outcomes. The results suggest that employees' attitude that covers both cognitive (i.e., functionality) and affective aspects could be a more impactful in predicting of important work criteria such as recruiting outcomes. The importance of this dimension is in line with TAM theory that emphasizes perceived usefulness (Davis, 1989; Venkatesh & Davis, 2000).

Notably, the contents of the AAAW scale and its dimensional structure correspond well with the expectancy-value model (Fishbein & Ajzen, 1975). According to the model, an individual's overall attitude is shaped by their subjective evaluations of attributes associated with the object and beliefs about the object. These attitudes predict behavioral intention and subsequent behaviors. In the context of AI applications, workers' subjective evaluations and beliefs associated with AI predict their intention to use AI technology and work outcomes. According to the expectancy-value model, there are prevailing accessible beliefs directly related to attitudes. In the AAAW scale, accessible beliefs encompass perceived humanlikeness, perceived adaptability, and perceived quality of AI. The remaining three attitudes such as personal utility, AI use anxiety, and job insecurity relate to expected values. Specifically, personal utility relates to utility (i.e., goal relevance), attainment (i.e., personal importance), and intrinsic values (i.e., enjoyment), and AI use anxiety and job insecurity reflect cost, which refers to fear of engaging in the task as well as lost opportunities that result from using the object (Eccles &Wigfield, 2002). While the expectancy-value model originally focused on the cognitive aspects of attitudes, researchers have extended its scope to include emotional process (Eccles &Wigfield, 2002), which aligns with the AAAW structure. Taking an inductive approach, our scale was not intended to validate a specific theory, but the contents and the scale structure are in accordance with the existing theory.

The emergence of the three functionality-related dimensions is also echoed by existing theories on technology acceptance and intelligent products that specify characteristics of technology (Davis, 1989; Rijsdijk et al., 2007; Wixom & Todd, 2005). Further, going beyond previous measures on new technology in general, the current research suggests that the inclusion of attitudes specific to AI such as perceived humanlikeness and perceived adaptability enhances an overall understanding of workers' attitudes towards AI application by providing more nuanced insights that are specific to work settings. For example, the association between perceived humanlikeness and conscientiousness was negative in Study 2, which contradicts previous research showing that consumers' anthropomorphism is *positively* correlated with conscientiousness (Letheren et al., 2016). Conscientious people's desire to gain mastery had a positive effect on anthropomorphizing consumer products (Letheren et al., 2016), but the same individuals might fear AI technology threats their desire to gain mastery, thus they less likely to perceive humanlikeness.

In the current study, perceived humanlikeness emerged as a unique dimension, but its effects were relatively weak and inconsistent. Despite increasing attention on anthropomorphism and humanlikeness of AI (Epley et al., 2007; Waytz et al., 2010), perceived humanlikeness exhibited the lowest ratings among the six dimensions and its effect was not statistically significant in predicting any of the three recruiting outcomes. We suspect that this may be because we did not specify physical appearance of AI technology; it has been suggested that the low visibility of embedded AI which tends to have less clear impact on a user's attitude (Glickson & Woolley, 2020). Future research should further investigate this possibility with more targeted study designs that distinguishes different types of AI application at work.

That said, one possibility worth exploring is whether perceived humanlikeness interacts with other AAAW dimensions in predicting work outcomes. For example, perceived humanlikeness could facilitate the positive effect of functionality-related attitudes on worker attitudes and outcomes because a heightened perception of humanlikeness indicates that the AI technology is capable of observing, thinking, and evaluating more effectively, serving as a source of normative social influence on the user (Waytz et al., 2010). When workers perceive higher levels of perceived humanlikeness, their functionality-related attitudes such as perceived adaptability and personal utility could be more positive. Using participants in the AI recruiting condition, we explored this possibility in a post-hoc analysis and found that the interaction between perceived adaptability and perceived humanlikeness was significant in predicting organizational attractiveness ($B = .15$, $p = .011$), intention to apply ($B = .20$, $p = .001$), and intention to recommend the organization ($B = .11$, $p = .087$). Consistently, perceived humanlikeness strengthened the positive effects of perceived adaptability on the recruiting outcomes. However, the interaction between personal utility and perceived humanlikeness on the recruiting outcomes was not significant (all $ps < .05$). The positive moderating role of perceived humanlikeness was limited to the unique features of AI, perceived adaptability. The results suggest that exploring potential interactions among these attitudes and their respective effects could enhance our understanding of workers' attitudes towards AI.

The current article contributes knowledge on the relationship between personality traits and attitudes towards AI application. The attitudinal dimensions of the AAAW were meaningfully associated with personality characteristics such as the Big Five personality traits, trust propensity, occupational self-efficacy and PIIT. Consistent with previous studies, our findings suggest that PIIT plays a generally positive role on diverse attitudes toward new technology including AI (Oostrom et al., 2013; Park & Woo, 2022). Further, the associations between the AAAW and other personality characteristics revealed more nuanced findings. For example, we found that perceived humanlikeness is positively related to trust propensity while negatively with occupational self-efficacy. Higher levels of PIIT and occupational self-efficacy seem to contribute to lower levels of AI use anxiety and job insecurity while trust propensity do not seem to have such effects. Trust in AI has been discussed as one of the critical factors in adopting AI (Glikson & Woolley, 2020; Schaefer, 2016). However, empirical studies on the effects of trust propensity on diverse attitudes towards AI are limited. The present findings indicate that the efficacious feelings and confidence based on knowledge and experiences using AI may complement the role of trust. Consequently, increasing efficacious feelings and confidence through learning and experiencing AI technology may be useful in shaping positive attitudes and preventing anxious attitudes towards AI application in the workplace.

The current research also contributes to the existing literature on the AI-based selection and recruitment literature. The uptake of AI technology conveys valuable information to (potential) job applicants. Importantly, our findings suggest that workers' interpretation of the signal depends on their existing attitudes towards AI application, rather than the level of AI application *per se.* In particular, we found that personal utility and AI use anxiety dimensions may be most influential in the recruiting context, whereas job insecurity may not be as relevant. Considering prior research on deleterious effects of job insecurity among workers (Brougham & Haar, 2018, 2018; Nam, 2019) as well as findings of the current study, we suggest that perceptions of AI being a threat to one's job security may be more detrimental to job incumbents than job seekers and applicants. This also implies that the effects of negative or positive attitudinal dimensions cannot be generalized across targets (i.e., employees, job applicants), and that it is important to consider which dimensions matter for a specific group of interest and how to prevent and promote specific attitudes within a given context.

Finally, our research shows that the six dimensions of AAAW are neither hierarchically nor sequentially structured. The literature on technology adoption often suggests sequential processes in which characteristics of technology precede individuals' attitudes toward new technology, which in turn affect one's actual usage of the technology (Davis, 1989; Venkatesh & Davis, 2000). However, the six dimensions in the current research did not exhibit any evidence of a causal or sequential relation. Because we excluded behavioral intention items during the item generation process, the AAAW may precede behavioral intention to use AI and behaviors, in line with our research (Davis, 1989; Venkatesh & Davis, 2000). Our findings suggest that how each dimension of AAAW functions in relation to job attitudes and outcomes seem to be more appropriate for using the scale rather than testing a sequential process within the six-factor structure.

Practical Implications

Organizations can use the AAAW measure to assess the effectiveness of their efforts to integrate AI technology into their work environment. The current findings suggest that reducing AI use anxiety and job insecurity among employees while increasing their perceptions of personal utility would help cultivate favorable attitudes towards AI application and subsequent changes in their behaviors and work experiences. This is particularly relevant when organizations begin to implement AI technology into selection system and job designs. For example, HR practitioners could gauge the extent to which current applications of AI implemented in their organizations are effective by surveying their employees' attitudes toward the AI technology, using the AAAW scale. Tracing changes in the AAAW dimensions can enable organizational leaders and HR managers to develop HR and training programs and evaluate the effectiveness of the programs.

In a series of supplementary analyses, we explored how demographic characteristics such as age, gender, and industry types were associated with the AAAW dimensions (see Appendix D for details). We found that age was negatively correlated with perceived humanlikeness and perceived adaptability, but not with the other attitudinal dimensions. Women, on average, reported higher levels of AI use anxiety and lower levels of personal utility than men. Those from information technology industries exhibited lower levels of AI use anxiety compared to others. These findings underscore the utility of the AAAW scale in assessing nuanced attitudes towards AI application, and provide insights into the targeted focus areas for organizational and educational efforts based on demographic considerations.

Our findings also indicate that AAAW predicts how job applicants' attitudes are impacted by AI use in recruitment, which have implications for employees, educators, and AI developers. In particular, holding negative attitudes towards AI seems to have a significant adverse impact on recruiting outcomes when the company is utilizing AI in their recruiting practices. Given that AI-enabled recruitment tools are increasingly popular, educators may want to target their efforts on addressing and mitigating students' (i.e., those constituting the future workforce) negative attitudes towards AI so that they are not disadvantaged during the recruiting process. Also, the fact that personal utility is related to PIIT and occupational self-efficacy (and predicts favorable recruitment outcomes) suggests that it may be beneficial for AI developers and employees to invest in educational and intervention programs to provide future workers (e.g., students, potential applicants) with more exposure to AI technology that leads to greater confidence, a sense of control, and feelings of enjoyment.

Limitations and Future Research Directions

The current research is subject to several limitations that suggest opportunities for future inquiry. One significant limitation of this study is to examine the criterion-related validity of the AAAW scale through the use of a hypothetical scenario. Although a hypothetical scenario is a valid method to gauge job applicants' attitudes toward organizations and recruiting methods, meta-analysis has revealed that job applicants' attitudes may be influenced by whether they are captured in the field versus a hypothetical and lab setting (Blacksmith et al., 2016). The negative effects of using digital selection methods on job applicants were more pronounced in the field than in a lab setting. We speculate that the AAAW dimensions may exhibit stronger effects in predicting job applicant reactions in a real AI recruitment context than in a hypothetical situation, which can be empirically examined in future studies. Also, in a field setting, exploring the impact of AI applications on other recruiting outcomes such as intention to accept the job offer could provide valuable insights for organizations and practitioners.

Apart from recruiting contexts, the AAAW scale's criterion-related validity remains a topic for future research. In line with technology acceptance theory, the AAAW dimensions are expected to have direct influences on behavioral intentions concerning AI such as the intention to use, accept, and collaborate with AI (e.g., Harris-Watson et al., 2023). AAAW is also likely to affect individuals' on-the-job behaviors and attitudes toward work and organizations where AI technology is utilized as suggested by previous studies (e.g., Brougham & Haar, 2018; Harris-Watson et al., 2023; Yam et al., 2023).

Another limitation is that the types of AI technology were not specified in our study. Although we believe the AAAW dimensions do apply to different types of AI technology such as embedded AI or virtual agents, the relative strengths of each dimension may differ in affecting outcomes as a function of AI types. For example, *perceived humanlikeness* could exhibit stronger effects in virtual agents than embedded AI due to more anthropomorphic features of virtual agents. The relative effects of the AAAW dimensions can also be examined with specific types of outcomes generated by AI. Individuals tend to accept outcomes of AI when the outcomes include low construal aspects (e.g., how to perform a task) rather than high construal features (e.g., why workers perform a task) (Kim & Duhachek, 2020). The dimensions that reflect unique AI features such as *perceived humanlikeness* and *perceived adaptability* may affect the willingness to accept outcomes of high construal features, whereas functionality-related attitudinal dimensions may be more strongly associated with AI-generated outcomes at low construal level. Thus, how AAAW interacts with AI developmental types and differential outcome would be an interesting avenue for future research.

Also, one may question the generalizability of our findings regarding the AAAW scale's validity. Although we recruited relatively large-sized samples and diverse occupations, all of the three studies were based on individuals in the United States. Cultural differences in perceptions and preferences of AI technology may exist (Awad et al., 2019). For instance, it has been suggested that East Asians are more likely to anthropomorphize objects including AI technology and thus tend to embrace AI more positively than Westerners (Kim & Duhachek, 2020). A large-scale study using participants from more than 200 countries demonstrate cultural differences toward moral decisions by AI (Awad et al., 2018). A country-level cultural orientation such as individualism and collectivism relates to moral decisions related to AI technology (Awad et al., 2018). Based on previous findings on the potential cross-cultural differences in how AI is perceived and preferred, future research should explore whether and how the dimensions of the AAAW are replicated across cultures.

Additionally, investigating how the AAAW dimensions change across time would be an interesting avenue to explore. For example, the emergence of AI use anxiety could reflect initial fear towards AI technology, which might decrease as AI becomes more integrated within organizations and society. However, according to a recent study, participants in less economically developed countries such as Africa and Central Asia tend to show lower levels of anxiety towards AI management than those in western developed countries such as America and Germany (Mantello et al., 2023), showing exposure to advanced technology does not always lead to positive perceptions of AI. Thus, we believe that the dimension remains an important dimension even in the future given the inherent uncertainty associated with AI technology. The topic warrants future research employing longitudinal designs.

Lastly, to extend the theory on the AAAW, future research that incorporates the AAAW and existing theories is needed. For example, whether trust in AI (e.g., Glikson & Woolley, 2020) is best understood as a construct that is distinct from (yet related to) AAAW or a subcomponent of AAAW (to be distinguished from the six AAAW dimensions identified from our study) needs future investigation. Trust may function as a mediator linking the dimensions of AAAW (e.g., perceived quality, humanlikeness) and ultimate acceptance/adoption of the technology (Glikson & Woolley, 2020). Also, given that trust develops from and is sustained through reciprocal processes and the level of trust varies depending on the focus (e.g., AI, coworker, organization), more experiences of using and collaborating with AI contribute to changes in the AAAW dimensions. Future investigation is warranted to examine whether “trust” serves as a key mediating mechanism that connects various dimensions of AAAW and actual usage of AI, and to update the content coverage of the AAAW.

Conclusion

Workers and employers are both experiencing unprecedented changes created by AI. Understanding workers’ attitudes towards AI application could be one of the initial steps to successfully adopting and utilizing AI technology for both organizations and their (potential and current) employees. In this research, we introduced an integrative framework for understanding and assessing worker’s attitudes towards AI in the workplace as a multidimensional construct. Across the three studies, the psychometric properties of the AAAW scale were evaluated and established. It is our hope that this research provides a useful reference and a measurement tool for future research on the interface between people and autonomous machines in the workplace.

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Table 1
Factor Loadings of the AAAW Items (Study 2)

	Factor loadings					
	1	2	3	4	5	6
Perceived humanlikeness of AI						
AI has desires.	**.93**	-.01	.00	-.02	.01	-.04
AI has beliefs.	**.87**	.03	-.02	-.02	.00	-.01
AI has ability to experience emotion.	**.84**	-.02	.02	.03	-.01	.04
AI has free will.	**.84**	.00	.01	.04	.01	.01
Perceived adaptability of AI						
AI learns from experience at work.	.00	**.94**	-.06	.01	.00	-.01
AI improves itself at work.	.01	**.88**	-.04	.00	-.01	.01
AI can learn at work.	-.01	**.82**	.04	.03	.01	.00
AI adapts itself over time at work.	-.01	**.70**	.12	-.02	.01	.00
Perceived quality of AI						
AI provides workers with a complete set of information.	.02	-.04	**.80**	.02	.00	-.04
AI produces correct information.	-.02	.04	**.74**	-.05	-.01	-.03
The information provided by AI is well formatted.	-.01	-.04	**.71**	-.03	.04	.04
AI operates reliably.	-.02	.06	**.67**	-.01	-.05	.06
The information from AI is always up to date.	.05	.01	**.66**	.06	.02	.02
AI use anxiety						
Using AI for work is somewhat intimidating to me.	-.03	.01	.00	**.87**	.00	.05
I would feel nervous operating AI in front of other people at work.	-.04	-.01	.03	**.86**	-.02	.09
I would feel uneasy if I was given a job where I had to use AI.	.03	.02	-.03	**.81**	-.01	-.09
I would feel paranoid talking with AI at work.	.08	.00	-.02	**.72**	.06	-.05
Job insecurity						
I am worried that what I can do now with my work skills will be replaced by AI.	.01	.02	-.03	-.05	**.94**	-.01
I am worried about my career due to AI replacing employees.	-.01	.00	-.02	.02	**.88**	.07
I think my job could be replaced by AI.	.05	.04	.00	-.05	**.80**	.05
I am worried about AI replacing what humans can do at work.	-.05	-.07	.07	.13	**.61**	-.13
Personal utility of AI						
Using AI would allow me to have increased confidence in my skills at work.	-.10	-.04	.02	.05	.04	**.91**

Using AI would provide me with personal feelings of worthwhile accomplishment at work.	.08	.00	-.01	.05	-.04	**.85**
Using AI would provide me with feelings of enjoyment at work from using the technology.	.01	.02	.03	-.02	.00	**.78**
Using AI would give me greater control over my work.	.05	.02	.02	-.08	.00	**.72**

Note. n = 1,069. AI = artificial intelligence; AAAW = attitudes towards AI at work. The highest loadings for each factor are bolded. This list represents the final items of the AAAW scale. This scale can be adapted to measure attitudes toward a specific AI technology (e.g., chatbot, automated video interviews) by replacing the word "AI" with the specific type of AI technology.

Table 2

Confirmatory Factor Analyses on the AAAW Items (Study 2)

	$\chi 2$	*df*	CFI	NFI	TLI	RMSEA	SRMR	AIC	BIC
Model 1: One-factor	12157.00	276	.33	.32	.33	.20	.21	12257.45	12498.75
Model 2: Five-factor (humanlikeness + adaptability)	3551.19	265	.81	.80	.79	.11	.13	.3671.19	3674.19
Model 3: Five-factor (AI use anxiety + job insecurity)	2674.34	265	.86	.85	.85	.09	.08	2794.34	3092.81
Model 4: Five-factor (quality + personal utility)	1686.53	265	.92	.91	.91	.06	.06	1806.53	2105.00
Model 5: One second-order factor	1359.04	269	.94	.92	.93	.06	.13	1471.04	1749.61
Model 6: Two second-order factors	1276.03	268	.94	.93	.94	.06	.13	1390.03	1673.58
Model 7: Uncorrelated six-factor	2185.91	275	.89	.88	.88	.08	.22	2285.91	2534.63
Model 8: Correlated six-factor	673.60	260	.98	.96	.97	.04	.04	803.60	1117.16

Note. *n* = 1,069. AAAW = attitudes towards AI at work; AIC= Akaike information criterion; BIC= Bayesian information criterion; CFI = comparative fit index; NFI = Bentler-Bonett normed fit index; RMSEA = root mean square error of approximation; SRMR = standardized root mean square residual; TLI=Tucker–Lewis index.

Table 3

Descriptive Statistics and Bivariate Correlations for Study Variables (Study 2)

	Variable	*M*	*SD*	1	2	3	4	5	6	7	8	9	10
1	AAAW: Humanlikeness	1.93	1.03	(.90)									
2	AAAW: Adaptability	3.72	0.84	.22**	(.91)								
3	AAAW: Quality	3.52	0.68	.29**	.45**	(.84)							
4	AAAW: AI use anxiety	2.54	1.03	.42**	-.07*	-.08**	(.89)						
5	AAAW: Job insecurity	2.90	1.08	.39**	.04	.12**	.43**	(.89)					
6	AAAW: Personal utility	3.11	0.96	.37**	.40**	.55**	-.19**	.03	(.90)				
7	Extraversion	3.07	0.89	.10**	-.01	.09**	-.11**	-.06*	.13**	(.76)			
8	Agreeableness	3.88	0.77	-.20**	-.06*	.09**	-.23**	-.14**	.02	.20**	(.74)		
9	Conscientiousness	3.97	0.79	-.29**	-.05	.03	-.30**	-.19**	-.05	.31**	.44**	(.78)	
10	Neuroticism	2.42	0.98	.12**	.01	-.07*	.30**	.21**	-.04	-.50**	-.41**	-.49**	(.84)
11	Openness	3.57	0.54	.03	.11**	.09**	-.04	.01	.07*	.24**	.23**	.15**	-.18**
12	Trust propensity	3.05	0.49	.25**	.18**	.25**	.04	-.01	.27**	.21**	.21**	.01	-.20**
13	Personal innovativeness in IT	3.57	0.81	-.03	.26**	.27**	-.36**	-.11**	.35**	.31**	.19**	.21**	-.26**
14	Occupational self-efficacy	4.10	0.57	-.12**	.11**	.17**	-.20**	-.14**	.10**	.37**	.34**	.49**	-.51**
15	Autonomy need	4.03	0.68	-.06	.11**	.19**	-.13**	-.05	.13**	.24**	.21**	.29**	-.29**
16	Competence need	4.02	0.73	-.06	.11**	.13**	-.10**	-.07*	.11**	.30**	.19**	.30**	-.29**
17	Relatedness need	3.87	0.86	.02	.05	.18**	-.04	-.05	.14**	.21**	.31**	.21**	-.21**
18	Self-actualization need	3.77	0.84	.11**	.07*	.24**	.02	.06*	.19**	.30**	.22**	.19**	-.25**
19	Physical striving need	3.52	0.89	.27**	.12**	.23**	.02	.06*	.24**	.31**	.09**	.15**	-.26**
20	Pleasure need	3.68	0.80	.16**	.12**	.22**	-.00	.04	.21**	.26**	.11**	.09**	-.15**
21	Money need	3.15	0.96	.30**	.12**	.22**	.09**	.06	.27**	.25**	-.01	.08*	-.18**
22	Security need	3.66	0.76	.15**	.09**	.23**	.08**	.05	.18**	.15**	.15**	.20**	-.17**
23	Self-esteem need	4.08	0.72	-.05	.09**	.17**	-.12**	-.07*	.11**	.35**	.28**	.33**	-.45**
24	Popularity need	3.40	0.91	.24**	.12**	.25**	.07*	.12**	.27**	.32**	.10**	.10**	-.18**

Note: n = 1,069. Values in the parentheses are Cronbach's alphas. AAAW = attitudes towards AI at work; AI = artificial intelligence; Personal innovativeness in IT = Personal innovativeness in information technology.
* $p < .05$, ** $p < .01$.

Table 3

Descriptive Statistics and Bivariate Correlations for Study Variables (Study 2) (Continued)

	Variable	11	12	13	14	15	16	17	18	19	20	21	22	23	24
1	AAAW: Humanlikeness														
2	AAAW: Adaptability														
3	AAAW: Quality														
4	AAAW: AI use anxiety														
5	AAAW: Job insecurity														
6	AAAW: Personal utility														
7	Extraversion														
8	Agreeableness														
9	Conscientiousness														
10	Neuroticism														
11	Openness	(.72)													
12	Trust propensity	.06	(.83)												
13	Personal innovativeness in IT	.22**	.13**	(.85)											
14	Occupational self-efficacy	.33**	.17**	.39**	(.76)										
15	Autonomy need	.32**	.11**	.23**	.45**	(.72)									
16	Competence need	.30**	.09**	.26**	.52**	.57**	(.75)								
17	Relatedness need	.23**	.21**	.19**	.32**	.43**	.27**	(.84)							
18	Self-actualization need	.34**	.17**	.21**	.37**	.60**	.51**	.45**	(.79)						
19	Physical striving need	.21**	.22**	.16**	.25**	.42**	.36**	.38**	.47**	(.78)					
20	Pleasure need	.28**	.15**	.21**	.27**	.49**	.36**	.41**	.57**	.55**	(.69)				
21	Money need	.11**	.21**	.18**	.23**	.30**	.32**	.28**	.34**	.43**	.40**	(.78)			
22	Security need	.14**	.20**	.15**	.31**	.41**	.40**	.35**	.43**	.47**	.38**	.47**	(.62)		
23	Self-esteem need	.28**	.18**	.24**	.53**	.62**	.63**	.42**	.59**	.47**	.45**	.35**	.42**	(.81)	
24	Popularity need	.20**	.24**	.23**	.28**	.35**	.39**	.45**	.47**	.41**	.42**	.48**	.40**	.43**	(.79)

Note: n = 1,069. Values in the parentheses are Cronbach's alphas. AAAW = attitudes towards AI at work; AI = artificial intelligence; Personal innovativeness in IT = Personal innovativeness in information technology.
* $p < .05$, ** $p < .01$.

Table 4

Moderating Effects of the AAAW Dimensions on Recruiting Outcomes (Study 3)

	Organizational attractiveness			Intention to apply			Intention to recommend		
Variable	*B*	*SE*	*t*	*B*	*SE*	*t*	*B*	*SE*	*t*
Step 1									
AI application level (A)	-0.07	0.05	-1.34	-0.01	0.05	-0.20	-0.06	0.05	-1.21
AAAW: Quality (B)	0.45	0.04	12.75**	0.38	0.04	10.18**	0.52	0.04	14.00**
R^2		.15			.10			.17	
Step 2									
AI application level (A)	-0.07	0.05	-1.34	-0.01	0.05	-0.20	-0.06	0.05	-1.21
AAAW: Quality (B)	0.36	0.05	7.18**	0.30	0.05	5.76**	0.42	0.05	8.04**
A X B	0.19	0.07	2.67**	0.15	0.07	2.08*	0.20	0.07	2.70**
R^2		.16			.10			.18	
Step 1									
AI application level (A)	-0.08	0.05	-1.66	-0.03	0.05	-0.52	-0.08	0.06	-1.48
AAAW: AI use anxiety (B)	-0.31	0.03	-10.85**	-0.30	0.03	-10.06	-0.30	0.03	-9.54**
R^2		.11			.10			.09	
Step 2									
AI application (A)	-0.08	0.05	-1.68	-0.03	0.05	-0.52	-0.08	0.06	-1.49
AAAW: AI use anxiety (B)	-0.23	0.04	-5.83**	-0.25	0.04	-6.10**	-0.27	0.04	-6.25**
A X B	-0.18	0.06	-3.13**	-0.11	0.06	-1.84	-0.07	0.06	-1.05
R^2		.12			.10			.09	
Step 1									
AI application level (A)	-0.08	0.05	-1.41	-0.02	0.06	-0.32	-0.07	0.06	-1.28
AAAW: Job insecurity (B)	-0.09	0.02	-3.65**	-0.06	0.02	-2.46*	-0.07	0.03	-3.49**
R^2		.01			.01			.01	
Step 2									
AI application (A)	-0.08	0.05	-1.41	-0.02	0.06	-0.32	-0.07	0.06	-1.28
AAAW: Job insecurity (B)	-0.05	0.03	-1.50	-0.03	0.03	-2.46	-0.05	0.03	-1.50
A X B	-0.08	0.05	-1.77	-0.07	0.05	-1.36	-0.08	0.05	-1.59
R^2		.02			.01			.01	
Step 1									
AI application level (A)	-0.08	0.05	-1.58	-0.02	0.05	-0.41	-0.08	0.05	-1.46
AAAW: Personal utility (B)	0.35	0.03	12.08**	0.30	0.03	10.00**	0.39	0.03	12.67**
R^2		.13			.09			.15	
Step 2									
AI application level (A)	-0.08	0.05	-1.59	-0.02	0.05	-0.41	-0.08	0.05	-1.48

AAAW: Personal utility (B)	0.27	0.04	7.01**	0.23	0.04	5.70**	0.29	0.04	7.03**
A X B	0.17	0.06	2.89**	0.15	0.06	2.54*	0.22	0.06	3.54**
R^2		.14			.10			.16	

Note. n = 942. AAAW = attitudes towards AI at work; AI = artificial intelligence
* $p < .05$, ** $p < .01$.

Table 5

Descriptive Statistics and Bivariate Correlations for Study Variables (Study 3)

		M	*SD*	1	2	3	4	5	6	7	8	9	10	11	12
1	AAAW: Humanlikeness	1.82	0.80	(.90)											
2	AAAW: Adaptability	3.60	0.94	.25**	(.92)										
3	AAAW: Quality	3.43	0.70	.17**	.30**	(.86)									
4	AAAW: AI use anxiety	2.49	0.88	.10**	-.25**	-.37**	(.88)								
5	AAAW: Job insecurity	2.46	1.13	.13**	-.09**	-.06	.41**	(.94)							
6	AAAW: Personal utility	3.15	0.87	.22**	.29**	.61**	-.52**	-.07*	(.87)						
7	Positive GAAIS	3.41	0.71	.08*	.38**	.50**	-.59**	-.17**	.72**	(.91)					
8	Negative GAAIS	2.59	0.79	.08**	-.19**	-.41**	.67**	.42**	-.45**	-.55**	(.87)				
9	NARS	2.91	0.72	-.09**	-.28**	-.39**	.74**	.27**	-.57**	-.62**	.66**	(.88)			
10	Job satisfaction	3.62	0.95	-.08*	.03	.12**	-.11**	-.17**	.05	.04	-.14**	-.07*	(.91)		
11	Organizational commitment	3.23	0.93	.02	-.02	.10**	-.05	-.17**	.09*	.06	-.11**	-.07	.74**	(.91)	
12	Life satisfaction	3.26	0.96	.03	-.03	.13**	-.06*	-.21**	.12**	.07*	-.14**	-.07*	.40**	.36**	(.90)

Note. n = 834–942. Values in the parentheses are Cronbach's alphas. AAAW = attitudes towards AI at work; AI = artificial intelligence; GAAIS = General Attitude toward Artificial Intelligence Scale; NARS = Negative Attitude toward Robot Scale.
* $p < .05$, ** $p < .01$.